

Raghunandan Kumar

+91 9555808473 · raghu9555k@gmail.com · [LinkedIn](#) · [GitHub](#) [Portfolio](#)

PROFESSIONAL SUMMARY

Full Stack & Agentic AI Engineer with 2+ years of hands-on experience building AI-powered and full-stack applications. Experienced in developing LLM pipelines, RAG systems, AI agents, semantic memory, and real-time applications using LangGraph, LangChain, Next.js, TypeScript, Python, and PostgreSQL. Strong experience in taking AI products from backend architecture and API integration to production-ready user interfaces.

EXPERIENCE

Ai Expert | Snorkel Ai (Contract)

July 2026 - Present Remote · San Francisco, California

- Rebuilt 7+ returned tasks from real merged open-source PRs across Python, TypeScript and Rust repos, taking each through a multi-stage automated review gate to acceptance.
- Runs oracle, base, forgery and idempotence scenarios directly from the packed submission zip, reproducing the grading platform's real environment. Caught defects that the automated gates scored as passing.
- An agent could score a perfect 1.0 by printing the graded test IDs without running a single test. Fixed with per-run secret tokens injected into test names at verify time, making the pass signal impossible to fake.
- The oracle solution was missing 13 of the PR's own test files, leaving repository tests failing while the reward still showed 1.0. Traced it to one root cause and restored full PR integrity with an automated audit.
- Reviewed other contributors' tasks and proved each finding in a container — including using git forensics (loose objects, missing remote, commit metadata) to show a shipped repo was a fresh git init with a fabricated base commit SHA.

Ai Engineer and Trainer | Outlier (Freelancer)

June 2026 - Present Remote · San Francisco, USA

- Evaluated LLM-generated responses for accuracy, reasoning quality, instruction following, relevance, and safety.
- Trained AI models through structured feedback, response correction, and comparative evaluation to improve reasoning and output quality.
- Designed and tested complex prompts to identify model weaknesses, edge cases, and failure patterns across different scenarios.
- Performed high-quality data annotation, labeling, and validation for AI training and evaluation datasets.
- Reviewed AI-generated solutions against detailed guidelines, identified quality issues, and provided clear justifications for ratings and corrections.

Software Engineering Intern (AI) | Tubo MI

Apr 2025 – May 2026 · Remote ·

- Engineered a Redis-backed scheduling engine using Sorted Sets and a purpose-built daemon, achieving sub-second reminder execution for workflows limited to a 24-hour window.
- Unified Swiggy food delivery and dining, Blinkit grocery services, and Google APIs within an AI-agent tool layer that supports contextual tool selection and user-driven routing.
- Developed a WhatsApp Business bot control interface with /help, /reset, and /new commands, while enabling environment-based runtime configuration to simplify deployment and initialization.
- Enhanced the WhatsApp agent pipeline with Azure Blob Storage-based file handling and support for managing message reactions.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++, SQL, Golang

Frontend: Next.js, React, Tailwind CSS, CodeMirror 6, HTML5 Canvas

Backend: Node.js, Express.js, FastAPI, Django, WebSocket, REST APIs, WhatsApp Business API

Agentic AI / LLM: LangChain, LangGraph, RAG Pipelines, Prompt Engineering, LLM API Integration Firecrawl, Inngest

Databases: PostgreSQL, Supabase (pgvector), MongoDB, Redis, Convex

Tools: Git, Jira, PyCharm, Docker, Azure (Blob Storage), AWS, GCP, Clerk, Sentry

PROJECTS

Mindly AI — Persistent Memory Agent | *Next.js 15, React 19, Groq, pgVector, Supabase, Redis, OAuth 2.0*

[Live](#) [Github](#) [Demo-Video](#)

- Engineered a dual-channel memory system using pgvector semantic search and PostgreSQL knowledge graphs for context-aware conversations across sessions.
- Built a four-stage embedding pipeline with automatic failover across Groq, Gemini, OpenRouter, and Ollama for reliable serverless execution.
- Designed a unified AI provider abstraction layer, making it easier to switch between multiple LLM providers without changing the core application logic.
- Implemented background processing for relationship extraction and memory updates, reducing user-facing latency while keeping conversational context up to date.

AI Interview Coach | *Next.js 16, Vapi AI SDK, Groq (Llama 3.3), Firebase, Face-API.js, TypeScript*

[Live](#) [Github](#) [Demo-Video](#)

- Integrated Vapi WebSocket SDK with Next.js to build low-latency AI voice interviews with real-time conversation and live transcription.
- Built a client-side face-tracking pipeline using Face-API.js to capture confidence and focus signals during interview sessions.
- Developed an automated evaluation pipeline using Groq (Llama 3.3), Vercel AI SDK, and Zod to generate structured, schema-validated interview feedback.
- Designed the end-to-end interview workflow with Firebase, connecting interview generation, voice sessions, transcripts, and AI-based performance evaluation.

Cognito — AI-Powered Browser IDE | *Next.js, WebContainer API, Claude Sonnet 4.6, TypeScript, Convex*

[Live](#) [Github](#)

- Built a browser-based full-stack IDE using WebContainer API and xterm.js, running complete Node.js environments client-side with zero server provisioning.
- Engineered custom CodeMirror 6 extensions for real-time LLM ghost-text autocomplete and context-aware Cmd+K code refactoring using Claude Sonnet 4.6.
- Developed an AI coding workflow that uses the active file and surrounding code context to generate relevant code suggestions and edits directly inside the editor.
- Integrated Convex for real-time project data and persistence, enabling users to manage files and continue AI-assisted development across sessions.

EDUCATION

B.Tech in Information Technology | **Guru Ghasidas Vishwavidyalaya**

Dec 2022 – Apr 2026 · Bilaspur, CG